

RESEARCH LETTER OPEN ACCESS

Unsupervised Machine Learning Identifies Asthma Phenotypes in the Population-Based West Sweden Asthma Study

Muwada Bashir Awad Bashir¹ | Daniil Lisik^{1,2}  | Saliha Selin Ozuygur Ermis¹  | Rani Basna³ | Reshed Abohalaka¹  | Selin Ercan¹ | Helena Backman²  | Teet Pullerits^{1,4} | Roxana Mincheva¹ | Göran Wennergren⁵  | Madeleine Rådinger¹  | Jan Lötvall¹ | Linda Ekerljung¹ | Hannu Kankaanranta^{1,6,7} | Bright I. Nwaru^{1,8}

¹Krefting Research Centre, Institute of Medicine, Sahlgrenska Academy, University of Gothenburg, Gothenburg, Sweden | ²Department of Public Health and Clinical Medicine/the OLIN Unit, Umeå University, Umeå, Sweden | ³Division of Geriatric Medicine, Department of Clinical Sciences in Malmö, Lund University, Malmö, Sweden | ⁴Section of Allergology, Sahlgrenska University Hospital, Gothenburg, Sweden | ⁵Department of Pediatrics, University of Gothenburg, Gothenburg, Sweden | ⁶Department of Respiratory Medicine, Seinäjoki Central Hospital, Seinäjoki, Finland | ⁷Faculty of Medicine and Health Technology, Tampere University Respiratory Research Group, Tampere University, Tampere, Finland | ⁸Wallenberg Centre for Molecular and Translational Medicine, Institute of Medicine, University of Gothenburg, Gothenburg, Sweden

Correspondence: Bright I. Nwaru (bright.nwaru@gu.se)

Received: 14 August 2024 | **Revised:** 4 July 2025 | **Accepted:** 3 October 2025

Funding: This work was supported by Hjärt-Lungfonden, 20210284, 244803506; Vetenskapsrådet, 2019-00247; Astma-och Allergiförbundet, F2021-0041; the Swedish state under the agreement between the Swedish Government and the county councils, Västra Götaland, ALFGBG-966075, ALFGBG-1005694; VBG Group Herman Krefting Foundation for Asthma and Allergy Research; NordForsk.

Keywords: adults | asthma | cluster analysis | deep learning | phenotypes | population-based study | Sweden | unsupervised clustering

To the Editor,

Asthma is a heterogeneous disease, and a clear-cut characterisation of its phenotypes has historically been daunting for both clinical practice and research purposes [1]. Asthma phenotypes have been defined primarily based on clinical experience [2]. Computational science is helping to elucidate biological processes, with ongoing applications in asthma phenotyping [3]. In this context, clustering algorithms can learn from unlabeled data to produce distinct subgroups. This data-driven approach is believed to be less subjective and, with relevant input data, can produce clinically meaningful phenotypes [4]. Characterising asthma at a more granular level aligns with efforts towards precision medicine, potentially enabling improved and tailored management. By including a broad range of clinical, biological, and epidemiological parameters, we employed a machine learning approach to identify and describe asthma phenotypes in adults.

The study sample was derived from the West Sweden Asthma Study (WSAS), a longitudinal cohort study investigating

different aspects of airway diseases among a representative sample of adults, which started in 2008 [5]. A total of 3101 individuals underwent clinical investigations, of which 1895 subjects who had ever had asthma were included in the current study. Asthma was defined as self-reported ever asthma. Age at onset was determined through the follow-up question “at what age did your asthma start?”. In total, 44 variables were selected, based on clinical experience and previous studies (see details at <https://osf.io/ucrnt>).

Missing data were imputed using multiple imputation with random forest. The phenotypes were derived using Deep Embedded Clustering, a novel approach that combines deep learning with clustering to discover patterns in complex data [6]. The R packages NbClust, Monte Carlo reference-based consensus clustering (M3C), and Adjusted Rand Index (ARI) were used to decide the optimal number of clusters based on consensus of 30 internal validation indices. The proposed numbers were evaluated in conjunction with clinical experience to determine the final optimal number of clusters (details at <https://osf.io/>)

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2025 The Author(s). *Clinical & Experimental Allergy* published by John Wiley & Sons Ltd.

Summary

- Several variables were subjected to a novel machine learning algorithm, deriving four distinct asthma phenotypes in adults.
- This study provides opportunity for profiling immunopathogenesis, therapeutic possibilities, and clinical outcomes of these phenotypes.

ucrnt/). Continuous data were expressed as means and standard deviations (SD). Group comparisons were performed by one-way analysis of variance with the Tukey post hoc test or Kruskal-Wallis test, as appropriate, for continuous variables and the Chi-square for categorical variables (details at <https://osf.io/ucrnt/>).

We determined that a four-cluster solution provided the best representation of the data:

- Cluster 1 ($n = 458$, 24.2%): Less-controlled, late-onset, non-atopic asthma with more females, and smokers
- Cluster 2 ($n = 545$, 28.7%): Female-dominated early adult-onset asthma

- Cluster 3 ($n = 358$, 18.9%): Adult-onset asthma with high T2 inflammation
- Cluster 4 ($n = 534$, 28.2%): Early-onset, mild, atopic asthma

Cluster 1 had the oldest average age at asthma onset (35 years), while cluster 4 had the youngest average age (13 years) (Figure 1). Cluster 1 also had the oldest average chronological age (65 years), highest average BMI (28.5 kg/m^2), highest proportion of smokers and average pack-year history, most respiratory symptoms, but the lowest proportion of allergic sensitization and family history of asthma or allergy. Cluster 1 also had the lowest post-bronchodilator FEV_1/FVC , highest neutrophil count, highest proportion of uncontrolled asthma based on Global Initiative for Asthma (GINA2021) classification, and second highest proportion of individuals who only used short-acting beta-agonists (SABA). Cluster 2 was the most female-dominated cluster and had the highest proportion of never smokers. Cluster 3 had the second oldest age at asthma onset, highest proportion of males, and highest FeNO levels and eosinophil count. Cluster 4 had the lowest average chronological age and BMI, equal distribution of males and females, lowest pack-years of smoking, least respiratory symptoms, highest allergic sensitization, highest FEV_1/FVC ratio, lowest reversibility, and lowest FeNO levels and neutrophil count. Furthermore, Cluster 4 had the lowest proportion of uncontrolled asthma, the lowest proportion of individuals on GINA

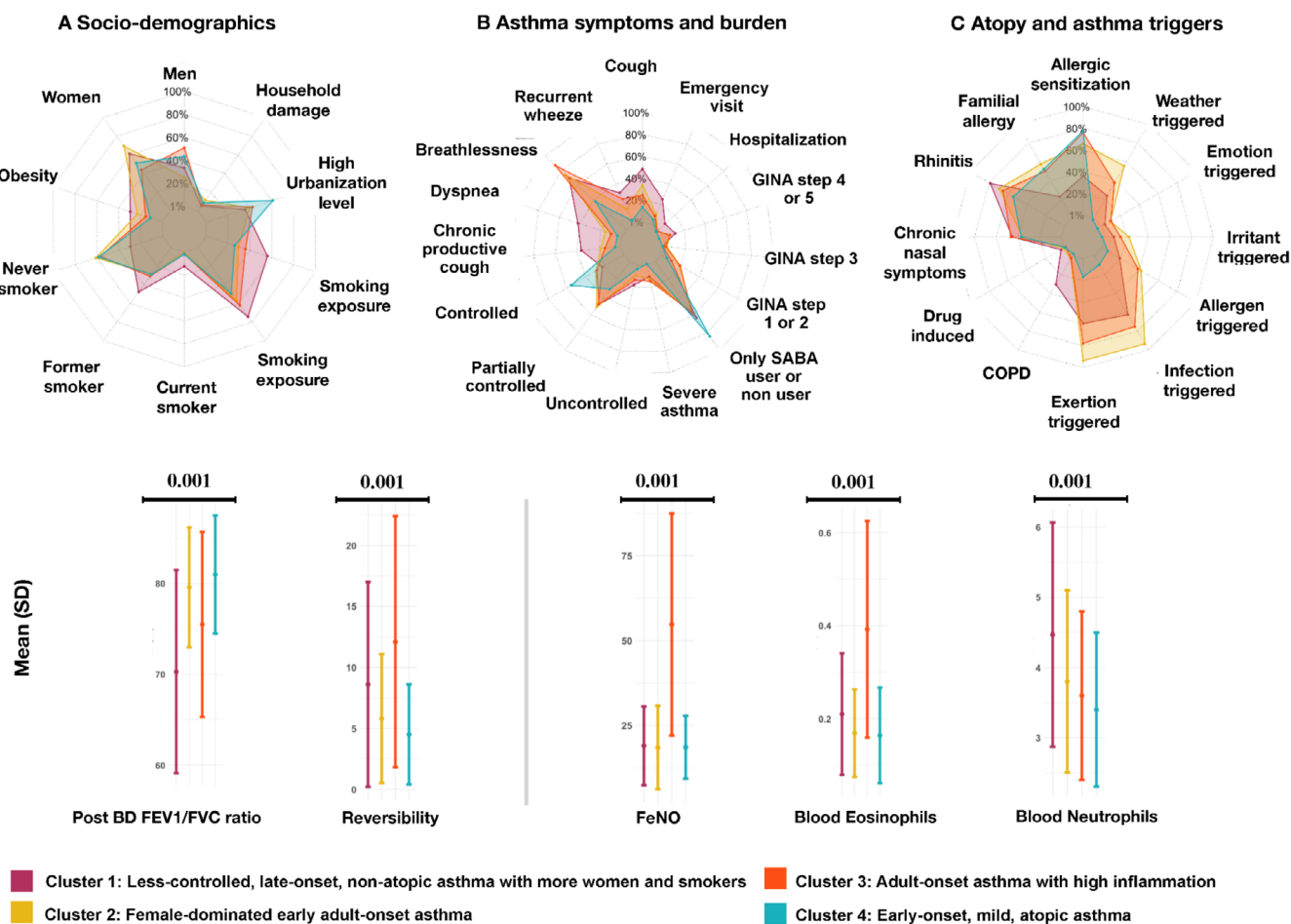


FIGURE 1 | Variations in demographic and clinical variables between derived asthma clusters.

treatment steps 4 and 5, and highest proportion of individuals who only used SABA (Figure 1).

Overall, the four clusters were distinguishable by age at onset, level of severity, symptom profiles, and measures of inflammation. Based on a population-based sample and a broad set of variables, a comprehensive characterisation of asthma severity as well as its associated clinical outcomes was achieved. The absence of certain inflammatory markers, like sputum measures, as well as data on certain co-morbidities may have limited our ability to fully capture relevant phenotypic features. Nevertheless, our findings closely align with previous studies [7, 8].

The asthma phenotype that typically begins in adulthood (Cluster 3) was characterized by significant inflammation and moderate symptom severity, a prominent feature being elevated FeNO and eosinophil counts. These are associated with type 2 immune response and may represent a group sensitive to inhaled corticosteroids therapy; however, they may be undertreated [9]. On the other hand, the early onset, mild atopic asthma phenotype (Cluster 4) probably represents the traditional childhood-onset asthma characterized by high allergic sensitization, better asthma control, and low symptom burden. Our data suggest that this phenotype may also have the highest rate of remission as it had the lowest proportion of those who have current asthma as defined by recent symptoms and medication use, indicating that although this phenotype develops during childhood, it might be transient.

In conclusion, four clinically relevant asthma phenotypes were identified in a representative sample of adults with ever asthma, distinguishable by age at asthma onset, severity, lung function, symptom patterns, and inflammation. These phenotypes provide a novel setting for catalyzing important asthma research, including detailed profiling of immunological pathways, modeling putative risk factors, characterizing comorbidity/multimorbidity profiles, and assessing future clinical outcomes.

Author Contributions

B.I.N. conceived the idea of the study. M.B.A.B. processed the data, performed the analysis, drafted the manuscript, and designed the figures under the supervision of B.I.N. L.E., G.W., H.K., and R.B. contributed to the implementation of the data processing and analysis. BN contributed to the design and implementation of the research and data collection. All authors discussed the results and contributed to the final manuscript. This study was supervised by B.I.N. and H.K.

Acknowledgements

We thank the participants for their valuable contributions to the study. We also thank the research nurses, Helén Törnqvist, Helen Friberg, Maryanne Raneklint, Lina Rönnebjerg, Louise Olausson, Victoria Malmberg, and Johanna Friberg, at Krefting Research Centre for their assistance with data collection. We are especially grateful to the late Professor Bo Lundbäck for all his work with the WSAS study.

Ethics Statement

The study was approved by the regional ethical review board in Gothenburg, Sweden. Ethics approval numbers 034–08, 906–16,

593–08, and 052–16. All participants gave written consent prior to participating in the study.

Conflicts of Interest

H.K. reports personal fees for lectures and consulting from AstraZeneca, Boehringer-Ingelheim, Chiesi, COVIS Pharma, GSK, Medscape, MSD, Novartis, Orion Pharma, and Sanofi. All other authors declare no conflicts of interest related to current work.

Data Availability Statement

The study data cannot be shared publicly due to an agreement with study participants and governance restrictions.

References

1. M. Deliu, T. S. Yavuz, M. Sperrin, et al., "Features of Asthma Which Provide Meaningful Insights for Understanding the Disease Heterogeneity," *Clinical and Experimental Allergy* 48, no. 1 (2018): 39–47.
2. M. L. Fajt and S. E. Wenzel, "Asthma Phenotypes in Adults and Clinical Implications," *Expert Review of Respiratory Medicine* 3, no. 6 (2009): 607–625.
3. D. Lefaudeux, B. De Meulder, M. J. Loza, et al., "U-BIOPRED Clinical Adult Asthma Clusters Linked to a Subset of Sputum Omics," *Journal of Allergy and Clinical Immunology* 139, no. 6 (2017): 1797–1807.
4. M. K. Ross, J. Yoon, A. van der Schaar, and M. van der Schaar, "Discovering Pediatric Asthma Phenotypes on the Basis of Response to Controller Medication Using Machine Learning," *Annals of the American Thoracic Society* 15, no. 1 (2018): 49–58.
5. B. I. Nwaru, L. Ekerljung, M. Rådinger, et al., "Cohort Profile: The West Sweden Asthma Study (WSAS): A Multidisciplinary Population-Based Longitudinal Study of Asthma,"
6. J. Xie, R. Girshick, and A. Farhadi, "Unsupervised Deep Embedding for Clustering Analysis," in *International Conference on Machine Learning* (Proceedings of Machine Learning Research, 2016), 478–487.
7. R. Nadif, M. Febrissy, M. Andrianjafimasy, et al., "Adult Asthma Phenotypes Identified by a Cluster Analysis on Clinical and Biological Characteristics," *European Respiratory Journal* 52, no. 2 (2018): OA293.
8. L. Wang, R. Liang, T. Zhou, et al., "Identification and Validation of Asthma Phenotypes in Chinese Population Using Cluster Analysis," *Annals of Allergy, Asthma & Immunology* 119, no. 4 (2017): 324–332.
9. M. J. Loza, R. Djukanovic, K. F. Chung, et al., "Validated and Longitudinally Stable Asthma Phenotypes Based on Cluster Analysis of the ADEPT Study," *Respiratory Research* 17 (2016): 21.